Will Lancer

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Education

University of Cambridge Oct. 2026 – June 2027 (expected)
Master of Advanced Study (MASt) in Theoretical Physics *Cambridge, UK*

Stony Brook University Aug. 2023 – May 2026
B.Sc. in Physics and Mathematics *Stony Brook, NY*

- **GPA:** 3.97 Overall, 3.99 Physics
- **Honors:** Goldwater Scholar, Sigma Pi Sigma (only sophomore inductee)

Research Manuscript

$h/2e$ **Flux Trapping without Superconductivity** 2026
Sole author; manuscript complete; awaiting advisor approval *Advisor: Zohar Komargodski*

- Developed an analytical account of magnetic flux trapping in ordinary metals, explaining how half a flux quantum ($h/2e$) can be trapped without electron pairing or a superconducting condensate
- Combined continuum calculations in disk and cylindrical geometries with lattice simulations to investigate finite-temperature effects and sensitivity to short-distance regularization

Experience

Undergraduate Reading Course and Research Sept. 2025 – Present
Simons Center for Geometry and Physics *Stony Brook, NY*

- Research on Aharonov–Bohm physics in metals with advisor Zohar Komargodski

Undergraduate ATLAS Researcher Jan. 2025 – Aug. 2025
Stony Brook University *Stony Brook, NY*

- Analysis focus: $t\bar{t}H$ with $H \to c\bar{c}$ (advisor: Hannah Arnold). Implemented charm-tagging preselection and event filters in ROOT/C++; validated MC samples.
- Analyzed kinematical data using C++ and machine-learning techniques; produced new selection method for charm quark jets; familiar with SALT flavor tagging.
- Ran batch jobs on large datasets; organized outputs and version-controlled analysis code for reproducibility.

Undergraduate Reading Course Oct. 2024 – May 2025
Stony Brook University *Stony Brook, NY*

- Studied Georgi's *Lie Algebras in Particle Physics* under Martin Roček: weekly problem sets/discussions on representations, Lie groups, and Lie algebras
- Audited PHY 610 (QFT I) under Alexander Zamolodchikov through reading course; studied path integrals, effective actions, RG flows, Wilson-Fisher fixed points, etc.

Coursework & Activities

- **Graduate Physics (all A):** Classical Mechanics; Classical Electrodynamics; Quantum Field Theory II; Group Theory for Theoretical Physics; Topology in Many-Body Physics and QFT; Advanced Statistical Mechanics
- **Mathematics:** Commutative Algebra and Algebraic Geometry (MAT 487; A); Real and Complex Analysis; Abstract Algebra; PDEs; proof-based Linear Algebra
- **Teaching:** Teaching Assistant for PHY 306 (Statistical Mechanics and Thermodynamics)
- **ML self-study:** Karpathy's *Zero to Hero* series; Kaplan's machine learning notes. **AI safety:** Readings in scalable oversight and AI control
- **Activities:** Society of Physics Students: wrote "Problem of the Week"; Undergraduate Mathematics Representative: helped preserve SBU's Math & Physics Library; Wrestling and Brazilian Jiu-Jitsu Clubs

Technical Skills and Awards

Programming Languages and Frameworks: C++, Python, PyTorch, Numpy, ROOT
Tools: Git, LaTeX, Monte Carlo workflows, ROOT-based data wrangling/ plotting
Additional Honors: Presidential Scholarship, Dean's List (every semester), URECA Summer Fellowship ($6k); URECA Researcher of the Month (Oct. 2025)